

A slow-weight obstruction to analytic renorming of uncountable Lorentz preduals

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Abstract

Bible asks in Remark 3.10 of [1] whether the uncountable Lorentz preduals $d_*(w, 1, A)$, already shown there to admit C^∞ -smooth renormings, admit analytic renormings. We prove a negative partial result. Under the standard monotone Lorentz-weight convention, if

$$w \notin \ell_q \quad \text{for every } 1 \leq q < \infty,$$

and A is uncountable, then $d_*(w, 1, A)$ admits no equivalent analytic norm. The arbitrary-weight case, in particular weights belonging to some finite ℓ_q , remains open here.

1 Source question

The source is V. Bible, *Using boundaries to find smooth norms*, arXiv:1311.1408. The relevant passage is Example 3.9 and Remark 3.10 on page 15:

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Example 3.9. $X = d_*(w, 1, A)$ has a C^∞ smooth equivalent renorming that ε -approximates the original norm.

Proof. Let $y \in S_X$. Since $\bar{y} \in c_0$, there exists $k \in \mathbb{N}$ such that $\bar{y}^k = 1$. It can also be shown $y \in c_0(A)$ and thus the supremum in the definition of $\bar{y}^k$ is attained. Following this, there exists $a_0, a_1, \dots, a_{k-1} \in A$ such that

$$1 = \bar{y}^k = \frac{\sum_{i=0}^{k-1} |y(a_i)|}{\sum_{i=0}^{k-1} w_i}.$$

Setting $\sigma = \{a_0, a_1, \dots, a_{k-1}\}$, we have $\|P_\sigma(y)\| = 1$. By Corollary 3.6, X has a C^∞ smooth equivalent renorming that ε -approximates the original norm. □

Remark 3.10. The space $X = d_*(w, 1, A)$ for A uncountable is a new example of a space with a C^∞ smooth renorming. It is not yet known if X has an analytic renorming.

Remark 3.11. In Theorem 2.1 and Corollary 3.1 we cannot drop the

Remark 3.10 says that, for uncountable A , $X = d_*(w, 1, A)$ is a new example of a space with a C^∞ -smooth renorming and that it is not yet known whether X has an analytic renorming.

2 Definitions and scope

Let $w = (w_j)_{j \geq 0}$ be a standard Lorentz weight: $w_0 = 1$, $w_j > 0$, $w_j \downarrow 0$, and $\sum_j w_j = \infty$. Put

$$W_k = \sum_{j=0}^{k-1} w_j \quad (k \geq 1).$$

For a set A , $d_*(w, 1, A)$ consists of all scalar functions x on A such that

$$\bar{x}_k = \sup \left\{ \frac{\sum_{i=0}^{k-1} |x(a_i)|}{W_k} : a_0, \dots, a_{k-1} \in A \text{ distinct} \right\}$$

defines a sequence $(\bar{x}_k) \in c_0$. The norm is $\|x\|_* = \sup_k \bar{x}_k$. Equivalently, if (x_i^*) is the decreasing rearrangement of the nonzero values of $|x|$, then $\|x\|_* = \sup_k (\sum_{i < k} x_i^*)/W_k$.

The additional hypothesis in this packet is

$$\sum_{j=0}^{\infty} w_j^q = \infty \quad \text{for every finite } q \geq 1. \quad (\text{SW})$$

This is a genuine restriction. It covers very slowly decreasing weights, for instance $w_j = 1/\log(j+2)$, but not weights such as $w_j = 1/(j+1)$.

3 Proof intuition

The proof isolates an obstruction to analytic norms that is familiar from $c_0(\omega_1)$, but adapted to the Lorentz-predual norm. Under (SW), continuous homogeneous polynomials on $d_*(w, 1, A)$ cannot see uncountably many coordinates. If an n -linear form had uncountably many nonzero coordinate coefficients, a Δ -system argument would produce disjoint coordinate patterns on which the form is uniformly nonzero. Feeding the form the vectors $\sum_j \pm w_j e_{a_j}$, whose $d_*(w, 1, A)$ -norms are bounded by one, and averaging signs makes the value dominate $\sum_j w_j^r$ for some $1 \leq r \leq n$. Hypothesis (SW) makes this impossible.

An analytic function is locally a countable sum of homogeneous polynomials. Thus, near the origin, it depends on only countably many coordinates. An analytic norm would yield an analytic separating function near the origin; a fresh coordinate outside that countable set gives the contradiction.

4 The partial result

Lemma 1 (Coordinate support of multilinear forms). *Assume (SW), and let A be uncountable. Let $T: X^n \rightarrow \mathbb{R}$ be a continuous n -linear form on $X = d_*(w, 1, A)$. Then there is a countable set $C \subseteq A$ such that*

$$T(e_{a_1}, \dots, e_{a_n}) = 0$$

whenever $\{a_1, \dots, a_n\} \not\subseteq C$.

Proof. Suppose not. By transfinite induction we can choose uncountably many ordered n -tuples $\mathbf{a}^\xi = (a_1^\xi, \dots, a_n^\xi)$, $\xi < \omega_1$, such that the finite set $F_\xi = \{a_1^\xi, \dots, a_n^\xi\}$ is not contained in the union of its predecessors and

$$T(e_{a_1^\xi}, \dots, e_{a_n^\xi}) \neq 0.$$

Passing to an uncountable subfamily, assume the absolute values are at least $\varepsilon > 0$, and that the equality pattern inside the ordered tuples is fixed. The Δ -system lemma gives a further uncountable subfamily for which the finite sets F_ξ have common root R , and the petals $F_\xi \setminus R$ are pairwise disjoint. The petals are nonempty by the choice of the tuples.

Let r be the number of argument positions occupied by petal coordinates, counting multiplicity. Thus $1 \leq r \leq n$. Choose any finite subfamily $\xi_0, \dots, \xi_m$. For root positions use the fixed unit vector corresponding to that root coordinate. For a petal position i , consider vectors of the form

$$x_i = \sum_{j=0}^m \eta_{i,j} w_j e_{a_i^{\xi_j}}, \quad |\eta_{i,j}| = 1.$$

Because the petal coordinates are distinct and w is decreasing, every such x_i has $\|x_i\|_* \leq 1$.

We choose the signs $\eta_{i,j}$ by the standard random-sign averaging trick: for each j , choose all but one petal-position signs independently and choose the last so that the product over the petal positions has the sign of $T(e_{a_1^{\xi_j}}, \dots, e_{a_n^{\xi_j}})$. Averaging over these signs, all mixed terms with petal indices not all equal have expectation zero, while the diagonal petal terms contribute positively. Hence some choice of signs satisfies

$$|T(x_1, \dots, x_n)| \geq \varepsilon \sum_{j=0}^m w_j^r.$$

The left hand side is bounded by $\|T\|$, since all root and petal input vectors have norm at most one. But (SW) implies $\sum_j w_j^r = \infty$, a contradiction as $m \rightarrow \infty$. $\square$

Corollary 1 (Coordinate support of polynomials). *Assume (SW), and let A be uncountable. Every continuous homogeneous polynomial on $d_*(w, 1, A)$ is countably supported: for each such polynomial P there is a countable $C \subseteq A$ such that*

$$P(x) = P(x \mathbf{1}_C) \quad (x \in d_*(w, 1, A)).$$

Proof. Apply Lemma 1 to the symmetric multilinear form associated to P , and use the polarization formula. $\square$

Fact 1 (analytic separation). *If a normed space admits an equivalent analytic norm, then there are $\rho > 0$ and an analytic function f , represented by a power series at the origin on a ball containing $\{x : \|x\| = \rho\}$, such that*

$$f(0) = 0 \quad \text{and} \quad \inf_{\|x\|=\rho} f(x) > 1.$$

This is the standard separating-function consequence of an analytic norm, used for example in the proof that no dense subspace of $c_0(\omega_1)$ admits an analytic norm in [2].

Theorem 1. *Let A be uncountable and let w satisfy (SW). Then*

$$X = d_*(w, 1, A)$$

does not admit an equivalent analytic norm.

Proof. Assume that X admits an equivalent analytic norm. Since equivalent norms give the same continuous polynomials, Corollary 1 applies to the homogeneous polynomials in the local power-series expansion of any analytic function for that norm.

Let f and ρ be as in Fact 1, and write the expansion of f at zero as

$$f(x) = \sum_{n=0}^{\infty} P_n(x)$$

on a ball containing the ρ -sphere. By Corollary 1, each P_n is supported on a countable set $C_n \subseteq A$. Let $C = \bigcup_n C_n$, still countable. Choose $a \in A \setminus C$ and take

$$x = \rho e_a / \|e_a\|$$

with respect to the analytic norm. Then x lies on the ρ -sphere, but $x \mathbf{1}_C = 0$. Therefore

$$f(x) = \sum_n P_n(x) = \sum_n P_n(0) = f(0) = 0,$$

contradicting $\inf_{\|y\|=\rho} f(y) > 1$. □

5 What remains open

The theorem does not settle Bible's Remark 3.10 for all Lorentz weights. The remaining case is when $w \in \ell_q$ for some finite q . In that regime the obstruction above is no longer available: diagonal high-degree polynomials such as $\sum_{a \in A} x(a)^q$ can become continuous, so analytic functions need not be countably supported.

6 Verification and novelty

The proof is symbolic and has no computational dependency. The source crop was generated from the arXiv PDF by `code/crop_open_problem.py`.

The run indexes were searched for 1311.1408, `Using boundaries to find smooth norms`, `d_*(w,1,A)`, `Lorentz predual`, and `analytic renorming`; no prior packet for this target was found. Web searches on 2026-06-20 for the exact Lorentz-predual analytic renorming question did not locate a later full answer. A nearby later paper [2] supplies the model analytic-support obstruction for $c_0(\omega_1)$, but it does not directly answer Bible's Banach-space question because the $d_*(w,1,A)$ -norm is stronger than the inherited $c_0(A)$ -norm.

7 Human review recommendation

Review should focus on the sign-averaging step in Lemma 1 and on the monotone-weight convention. If the source permits non-monotone weights literally, the packet should be read under the standard Lorentz replacement by the nonincreasing rearrangement of w , or the theorem statement should keep the explicit monotonicity assumption.

References

- [1] V. Bible, *Using boundaries to find smooth norms*, arXiv:1311.1408, 2013.
- [2] S. Dantas, P. Hájek, and T. Russo, *Smooth norms in dense subspaces of Banach spaces*, arXiv:1911.09611, 2019.